Collectively Big, Locally Small: Online Supplement

Systematic search, simulation study and coded corpus

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This supplement accompanies the survey *Collectively Big, Locally Small*. Section S1 documents the systematic database search summarized in Section 5.4 of the paper: sources, query strings, screening rules and their verification, the recall of the search against the coded corpus, the validation of the dictionary coder, and the field-level map. Section S2 gives the full design and results of the fourteen Monte Carlo simulations of the Graph Hierarchical Borrowing (GHB) blueprint summarized in Section 8.7. Section S3 lists the 290 coded works. Every number in this supplement is generated by the scripts released with the paper (`prisma/` and `simulations/`) and read into the text as LaTeX macros.

S1 SYSTEMATIC SEARCH

S1.1 Sources and query strings

The search was run on 8 October 2026 for records published from 1 January 2010 to the search date. PubMed was queried through the NCBI E-utilities with every term restricted to title and abstract (`[tiab]`). medRxiv and the arXiv records indexed by Europe PMC were queried through the Europe PMC REST API (source `PPR`, field `TITLE_ABS`). Europe PMC indexes only the life-science categories of arXiv. A direct search of the arXiv API was attempted but abandoned because the API rejected the full query; computer-science preprints are therefore covered only when they were also published in an indexed venue. Every query string is stored verbatim in `prisma/data/search_log.json` together with the hit counts; the PubMed string was

```
("small data"[tiab] OR "small sample*"[tiab] OR "limited data"[tiab] OR "data scarcity"[tiab] OR
"few-shot"[tiab] OR "low-resource"[tiab] OR "rare disease*"[tiab] OR "label-efficient"[tiab] OR
"data-efficient"[tiab] OR federated[tiab] OR decentralized[tiab] OR decentralised[tiab] OR
"privacy-preserving"[tiab]) AND (health*[tiab] OR clinical[tiab] OR medic*[tiab] OR
hospital*[tiab] OR patient*[tiab]) AND ("machine learning"[tiab] OR "deep learning"[tiab] OR
"transfer learning"[tiab] OR "synthetic data"[tiab] OR "data augmentation"[tiab] OR "active
learning"[tiab] OR "semi-supervised"[tiab] OR "self-supervised"[tiab] OR "foundation
model*"[tiab] OR "hierarchical model*"[tiab]) AND 2010/01/01:2026/10/08[dp]
```

Scopus, Web of Science, IEEE Xplore and the ACM Digital Library were not searched, because they require a subscription or an API key. Exports from them (RIS, BibTeX or CSV) can be added with `prisma/import_exports.py`; they are then deduplicated and screened with the other records, and the flow diagram and every number are regenerated.

S1.2 Screening rules

Records were deduplicated by DOI and by normalized title, keeping the PubMed version when a preprint had also been published. Titles and abstracts were then screened by the rules of Table S1, applied in order. The rules operationalize criteria I1, I2 and X1 to X3 of Section 5: they keep every record in which a scarcity term of the query refers to scarce, fragmented or poorly labeled data and a learning method is present, and they remove the frequent uses of the query terms that do not refer to data scarcity.

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Table S1. Screening rules, applied in order; the first rule that applies decides.

Rule	Excludes records whose	*n*
A1	abstract is missing, or language is not English	46
A2	publication type is editorial, letter, comment, erratum, guideline, consensus statement, protocol, audio-visual material or historical article, or which are retracted (X1)	54
R1	only data-scarcity term refers to a resource-limited setting ("low-resource settings, countries, hospitals")	421
R2	only data-scarcity term is "small sample" in a sentence on limitations or future work	264
R3	only decentralization term refers to decentralized clinical trials or care delivery	14
R4	title is a reporting guideline, checklist, statement, position paper, policy or consensus (X1)	8
R5	title and abstract contain no learning method	72
	Eligible on title and abstract	5,204

S1.3 Verification by a single reviewer

A single reviewer judged two random samples of 100 records each against criteria I1, I2 and X1 to X3, reading title and abstract; each sample drew 60 or 40 records from the included side and as many from the excluded side of the rules then in use. The first sample led to the removal of one rule, which excluded records whose scarcity term appears only in the author keywords (4 of its 8 false exclusions came from it). A rule added after the first sample, which excluded "limited data" in the sense of a gap in clinical knowledge, removed small-training-set studies in 3 of the 7 sampled records it excluded in the second sample and was also removed. Against all 200 judgments, the final rules have a precision of 63% and a negative predictive value of 94%; the sensitivity within the samples is 94%, but the samples are balanced by design, so this is not the sensitivity in the full set. A stricter rule, which required the scarcity term in the title or in a sentence that also mentions data, labels, training or a model, raised precision to 73% but lowered sensitivity to 77% and was not adopted. The judgments are released in `prisma/data/verify_judgments.json`.

S1.4 Recall against the coded corpus

A search is usually validated by its relative recall against a reference set of known relevant studies. Our reference set is the coded corpus: 201 of its works were published since 2010 and evaluated on health data, 144 of these are indexed in the searched sources (found in Europe PMC by DOI or title), and the protocol query retrieved 31 of them (22%). To test whether a better query would have found them, we split the 140 PubMed-indexed works at random into two halves and evaluated broadened variants of the query on PubMed (Table S2). Adding 39 terms to the scarcity block (sample size, multi-site, external validation, generalizability, shift, privacy, annotation and related terms) and 16 terms to the method block (artificial intelligence, neural networks, prediction models and related terms) multiplied the number of records by thirteen and raised recall to about one half. The literature on learning from scarce health data is not well delimited by vocabulary. A large part of it reports a method or a failure without naming scarcity at all, which is why the paper's synthesis rests on the snowballed corpus.

S1.5 Dictionary coder and field-level map

Each eligible record was coded from its title and abstract by dictionaries for the six provenance channels, the ten scarcity classes and the data modality (`prisma/classify.py`). The primary channel is the channel whose terms match most often, with title matches counted twice. On the 147 corpus works whose abstracts are available in Europe PMC, the dictionary coder agreed with the manual primary channel in 72% of cases (Cohen's $\kappa = 0.65$) and recovered 63% of the manually

Table S2. Number of PubMed records and relative recall on two random halves of the PubMed-indexed corpus works, for the protocol query and for broadened variants.

Query	PubMed records	Recall, first half	Recall, second half
Protocol query (Section 5.4)	5,885	17%	27%
Scarcity block broadened	40,584	39%	44%
Method block broadened	11,178	21%	31%
Both broadened	75,983	51%	54%

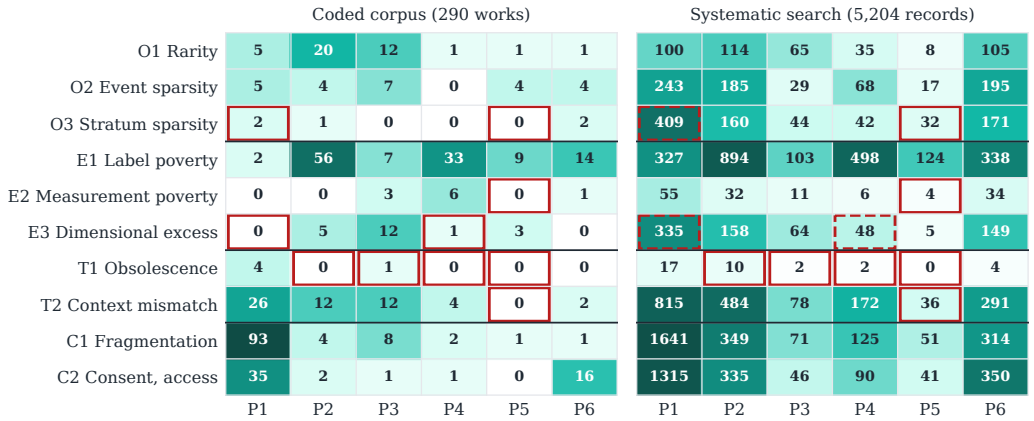


Fig. S1. Left: works of the coded corpus targeting each scarcity class through each channel, as in Figure 8 of the paper; red outlines mark the ten gap cells. Right: records of the systematic search coded by dictionary, shaded by percentile rank among the 60 cells. Solid outlines: gaps confirmed (bottom third of the field-level map); dashed outlines: gaps not confirmed.

assigned scarcity classes. Figure S1 sets the coverage of the coded corpus (Figure 8 of the paper) next to the field-level map. Because the dictionaries for stratum sparsity (which includes the term bias) and for dimensional excess (which includes omics terms) are broad, their rows are inflated in the field-level map; Section 5.4 of the paper discusses the three gap cells that the search does not confirm.

S2 SIMULATION STUDY OF THE BLUEPRINT

Section 8.7 of the paper summarizes these simulations; this section gives the full design and every figure. The simulations test the blueprint of Section 8 under its own model and under a logistic analogue. They check internal consistency and expose the cost of estimating the unknown quantities that Propositions 8.1 to 8.4 take as known; they are not evidence about real hospital networks. Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 and clipped (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, clipping bound $C = 1$, and the privatized releases are averaged by 60 rounds of Metropolis-Hastings gossip on the ring with two-hop links of Figure 10. *Plug-in GHB* estimates everything it needs from the privatized releases and public quantities only: τ^2 by DerSimonian-Laird (Remark 8.3), the leave-one-out population estimate by inverse-variance weighting, and its error M from those weights. Risk is reported relative to local estimation, R_k/v_k with $v_k = \sigma^2/n_k$; values below 1 mean

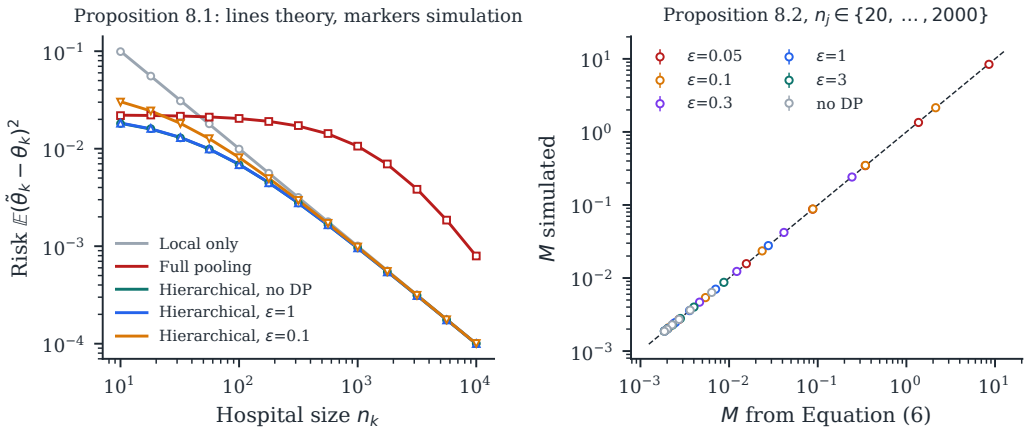


Fig. S2. Simulation 1. Left: risk against hospital size; lines are the closed forms of Figure 9, markers are simulated (±2 standard errors). Right: simulated M of the privatized leave-one-out mean against Equation (6), for 11 other hospitals of 20 to 2,000 patients and six privacy budgets.

that a hospital gains. All code is in `simulations/`, and `run_all.py` regenerates every figure and number in about two minutes.

Simulation 1: the closed forms. Figure S2 repeats the setting of Figure 9 with 40,000 replicates per point. Across 101 comparisons of a simulated risk or M with its closed form, 98% lie within two Monte Carlo standard errors, and the largest relative deviation is 1.6%. Propositions 8.1 and 8.2 and the code that draws Figure 9 agree.

Simulation 2: heterogeneity. Figure S3 varies τ^2 from 10^{-4} to 1. Plug-in GHB is never worse than local estimation by more than Monte Carlo error: its worst hospital reaches a relative risk of 1.02 at most. A single global model (the sample-size weighted mean of all site estimates, computed without privacy) is the best choice when τ^2 is tiny, but its mean relative risk reaches 134 and its worst hospital 382 at $\tau^2 = 1$. Estimating τ^2 costs most when heterogeneity is smallest: at $\tau^2 = 10^{-4}$ and $\epsilon = 1$ the plug-in reaches 0.12 against 0.07 with the true τ^2. Gossip with 60 rounds is indistinguishable from exact averaging.

Simulation 3: who gains, and how many hospitals are needed. At $\tau^2 = 0.02$ and $\epsilon = 1$ (Figure S4, left), every hospital gains about the same number of virtual patients, $B \approx 46$, as Equation (5) predicts. The hospitals with 60 patients or fewer multiply their effective sample size by 2.28 on average, and those with 300 or more by 1.08; the smallest gain of any hospital is 1.05. The global model multiplies the effective sample size of small hospitals by 1.44 but cuts that of large hospitals to 0.15 of their own. With few hospitals, τ^2 is poorly estimated (Figure S4, right). With three hospitals the mean relative risk is 0.91 with DerSimonian-Laird against 0.78 with the true τ^2, and the gap closes as the network grows (0.68 against 0.66 with 48 hospitals). The posterior mean of τ^2 under a half-Cauchy(0.5) prior on τ, the weakly informative choice that Remark 8.3 recommends, does better than DerSimonian-Laird from four hospitals upward but not with three.

Simulation 4: gossip rounds. Figure S5 shows the mean relative risk of plug-in GHB after T rounds of gossip on the four topologies of Figure 10. The risk comes within 2% of its exact-average value after 35 rounds on the ring, 22 on the regional clusters, 11 on the sparse random graph and 14 on

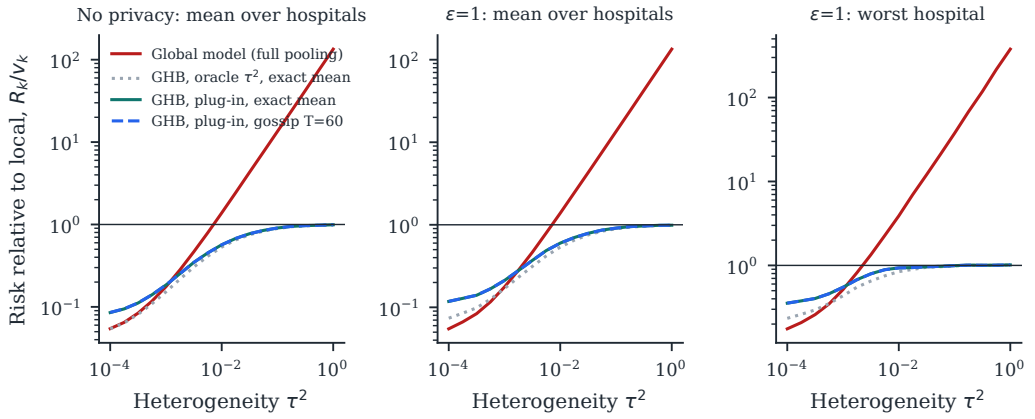


Fig. S3. Simulation 2. Risk relative to local estimation as a function of heterogeneity, mean over hospitals without privacy and with $\epsilon = 1$, and for the worst hospital with $\epsilon = 1$.

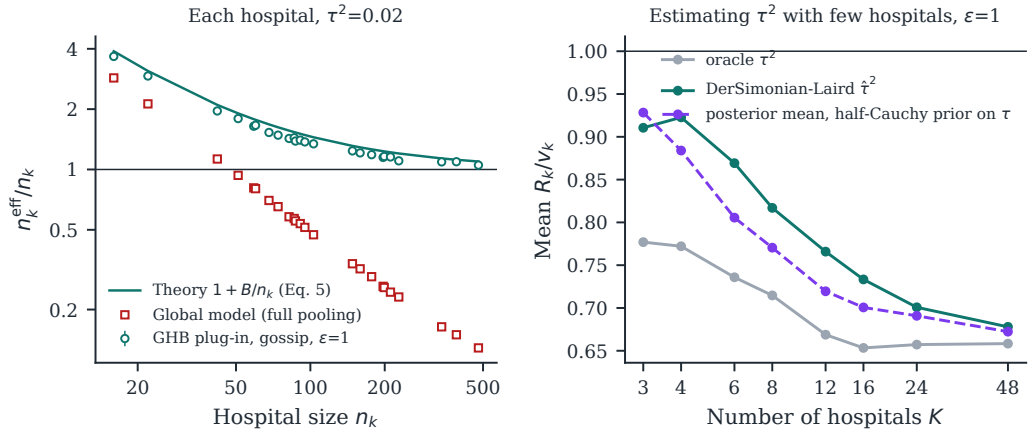


Fig. S4. Simulation 3. Left: effective sample size relative to own size for each hospital, plug-in GHB and a global model; the line is $1 + B/n_k$. Right: mean relative risk against the number of hospitals for three ways of obtaining τ^2.

the ring with two-hop links, far fewer than the rounds that the worst-case bound of Proposition 8.4 requires for a consensus error of 1% (203, 127, 73 and 69). The bound is conservative for the risk: it controls the consensus error of the releases, while the risk depends on that error only through a population estimate that the shrinkage weight already discounts. After a single round on the ring, GHB is worse than local estimation (relative risk 1.13), so a hospital should not use the borrowed estimate before the gossip has mixed.

Simulation 5: joint release by small hospitals. Section 8.3 proposes that small hospitals release jointly through secure aggregation, so that the privacy noise is calibrated to the group. Figure S6 shows that the remedy matters only when every hospital is small. In a network of 16 hospitals of 20 patients and 8 of 200, the large hospitals dominate the inverse-variance weighted population

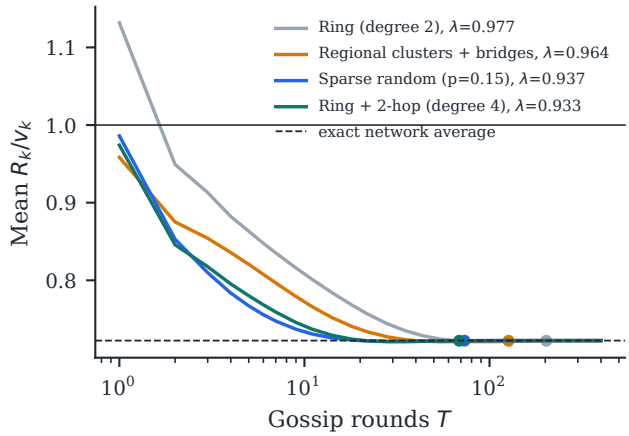


Fig. S5. Simulation 4. Mean relative risk of plug-in GHB against the number of gossip rounds ($\tau^2 = 0.02$, $\epsilon = 1$). Dots mark $\log(100)/(1-\lambda)$; the dashed line is the exact network average.

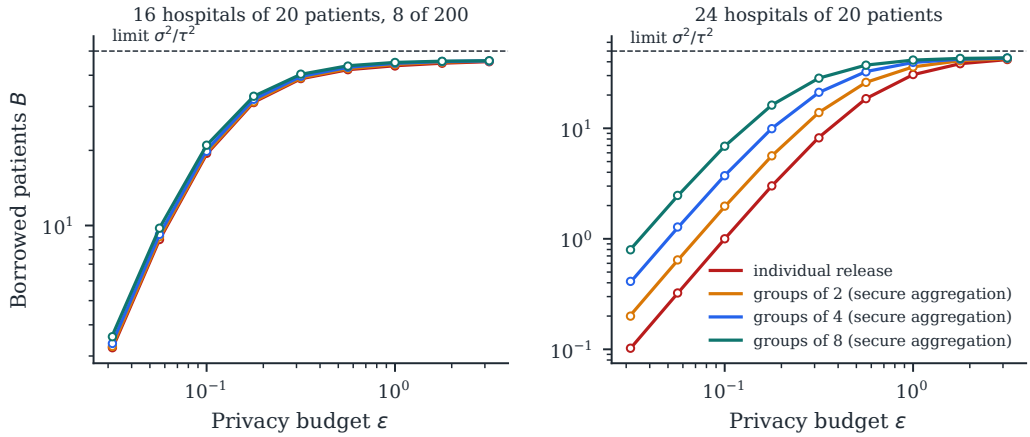


Fig. S6. Simulation 5. Borrowed patients B against the privacy budget for individual and joint release; lines are closed forms, markers simulation.

estimate, and groups of eight raise the borrowed patients only from 19.3 to 21.0 at $\epsilon = 0.1$. In a network of 24 hospitals of 20 patients, the same grouping raises B from 1.0 to 6.9 at $\epsilon = 0.1$ and from 30.6 to 41.4 at $\epsilon = 1$.

Simulation 6: structured heterogeneity. The 24 hospitals are split into four regions of six; θ_k is the sum of a regional effect with variance τ_r^2 and a hospital effect with variance $\tau_w^2 = 0.002$. The GMRF prior of Equation (7) links hospitals of the same region and is computed by Jacobi iterations on the privatized releases, followed by each hospital's local update with its own noise-free estimate. Its anchor term is switched off so that it uses the same information as the exchangeable prior, and its smoothing weight β was chosen on separate tuning replicates (the same value, $\beta \approx 46$, was selected at every τ_r^2); the regions are assumed known. Figure S7 shows a crossover. When regions barely

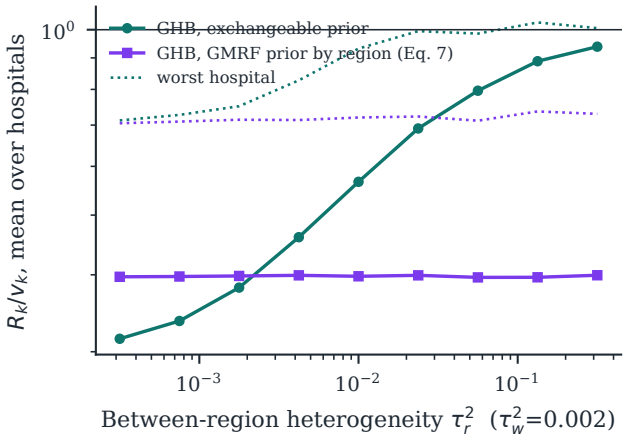


Fig. S7. Simulation 6. Mean relative risk (solid) and worst hospital (dotted) for the exchangeable and the regional GMRF prior, against the between-region variance ($\epsilon = 1$).

differ ($\tau_r^2 \lesssim 0.003$), the exchangeable prior is better (0.31 against 0.40 at the smallest τ_r^2), because the GMRF borrows only from five neighbors. When regional differences dominate, the exchangeable prior loses its gain (0.94) while the GMRF keeps it (0.40).

Simulation 7: logistic-regression heads. This simulation leaves the Gaussian model. Each hospital has its own logistic model on four standardized features and an intercept, $\theta_k = \mu + u_k$ with $u_k \sim \mathcal{N}(0, \tau^2 I)$, and features are clipped so that $\|x\| \le \sqrt{5}$. Each hospital releases the minimizer of the ridge objective $\frac{1}{n}\sum_i \ell_i(\theta) + \frac{\lambda}{2}\|\theta - c\|^2$ centred on a fixed public reference $c = 0$, with $\lambda = 0.2$, whose ℓ_2 sensitivity is $2\sqrt{5}/(n\lambda)$ whatever the centre, plus isotropic Gaussian noise (output perturbation). The ridge shrinkage is then undone to first order with the Fisher information at c under the public feature distribution; this is post-processing. Here μ_0 is a published model (channel P3), equal to μ plus noise of standard deviation 0.3; it enters only through the prior of each hospital's final fit. GHB borrows coordinatewise from these releases, with τ^2 from its half-Cauchy posterior and the network estimate combined with μ_0 by precision weighting (layer L3), and fits each hospital's model by maximum a posteriori with the resulting Gaussian prior. The baselines are a local model with prior $\mathcal{N}(\mu_0, I)$, which uses the anchor but no other hospital, and a global model fitted on the pooled data of all hospitals without privacy. Table S3 and Figure S8 report the excess log-loss over the true model on 4,000 new patients per hospital, averaged over 100 replicates. At $\tau^2 = 0.05$ and $\epsilon = 1$, GHB reduces the excess log-loss of local estimation by 39% (0.0266 to 0.0161), below the global model trained without privacy (0.0173), and no hospital is worse off on average (largest ratio to local 0.91, against 2.48 for the global model). Without privacy GHB reaches 0.0133, better than the global model; output perturbation is therefore the main cost at $\epsilon = 1$. At $\epsilon = 0.3$ only a small gain remains (0.0261 against 0.0273).

Development note. Three components of this simulation were added after the first runs failed. The first version used the ridge releases without undoing the shrinkage, estimated τ^2 by DerSimonian-Laird and ignored μ_0. With privacy noise at $\epsilon = 1$ its estimates of τ^2 were erratic (median 0, mean several times the truth), and it was worse than local estimation at $\tau^2 = 0.2$ and at $\epsilon = 0.3$. Each component follows the design of the paper (post-processing, Remark 8.3 and layer L3), but they were chosen with knowledge of these failures, and the results should be read accordingly. A fourth

Table S3. Simulation 7. Excess log-loss over the true model (nats; lowest of the three methods in bold) and AUC, mean over 24 hospitals and 100 replicates. Oracle: the true model of each hospital. Global: one model on the pooled data, without privacy.

		Excess log-loss			AUC			
τ^2	ϵ	Local	Global	GHB	Local	Global	GHB	Oracle
0.01	1	0.0283	**0.0041**	0.0157	0.715	0.733	0.722	0.737
0.05	1	0.0266	0.0173	**0.0161**	0.723	0.728	0.728	0.743
0.2	1	0.0268	0.0601	**0.0205**	0.749	0.717	0.752	0.770
0.5	1	**0.0271**	0.1257	0.0281	0.784	0.688	0.782	0.804
0.05	0.3	0.0273	**0.0167**	0.0261	0.725	0.730	0.725	0.745
0.05	3	0.0271	0.0169	**0.0124**	0.722	0.728	0.732	0.743
0.05	10	0.0282	0.0170	**0.0130**	0.723	0.729	0.734	0.743
0.01	∞	0.0274	**0.0042**	0.0075	0.717	0.734	0.734	0.737
0.05	∞	0.0281	0.0170	**0.0133**	0.722	0.729	0.734	0.743
0.2	∞	0.0274	0.0599	**0.0222**	0.745	0.712	0.747	0.766
0.5	∞	**0.0277**	0.1259	0.0289	0.785	0.690	0.784	0.805

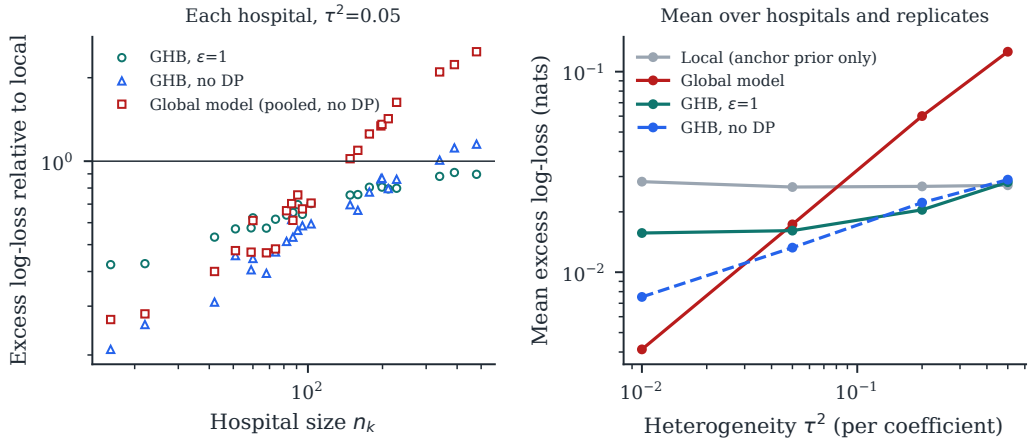


Fig. S8. Simulation 7. Left: excess log-loss relative to local estimation for each hospital ($\tau^2 = 0.05$). Right: mean excess log-loss against heterogeneity.

change came from Simulation 11: the releases were first centred on the anchor ($c = \mu_0$), and the shrinkage was undone with the Fisher information at μ_0. When the anchor was badly wrong, this corrupted every release, and GHB without the anchor reached an excess log-loss of 0.34 against 0.06 for local estimation at an anchor error of two standard deviations. Centring on a fixed reference removes the dependence; Simulations 7, 8, 10 and 11 are reported with it. With anchor-centred releases, Simulation 7 had given a slightly smaller gain at $\epsilon = 1$ and no gain at $\epsilon = 0.3$. The first version of Simulation 6 also gave the GMRF prior the anchor μ_0, which the exchangeable prior did not have; the version reported here removes it.

Simulation 8: federated and meta-analytic baselines. The logistic world of Simulation 7, at $\epsilon = 1$ and four levels of heterogeneity. All private methods use the same releases, so they satisfy the same guarantee: one-shot federated averaging (the sample-size weighted mean of the releases, one

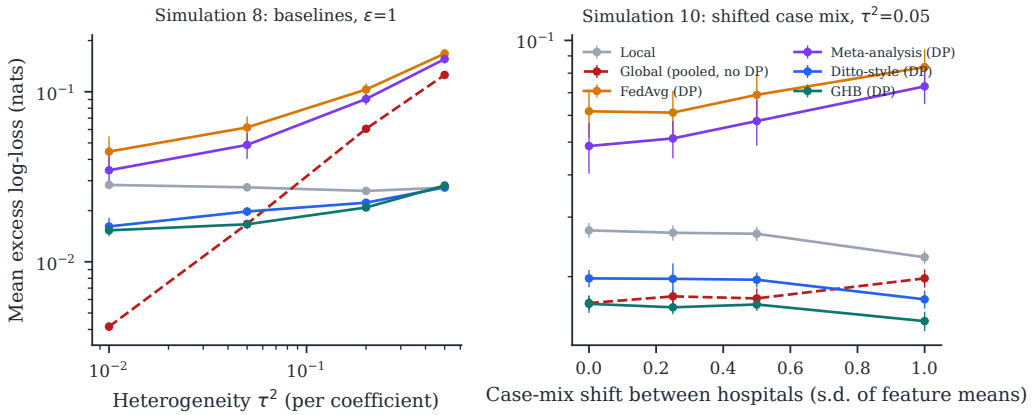


Fig. S9. Simulations 8 and 10. Mean excess log-loss of GHB and of private and non-private baselines against heterogeneity (left) and case-mix shift (right). Bars: ±2 standard errors.

model for every hospital), the random-effects meta-analytic mean (DerSimonian-Laird weights, one model for every hospital), and a Ditto-style personalized model, which fits each hospital's data with a proximal term $\frac{\lambda}{2}\|\theta - \tilde{\theta}_{\text{FedAvg}}\|^2$. The weight λ is chosen from $\{0.3, 1, 3, 10, 30, 100\}$ on ten separate tuning replicates per setting, which favours Ditto-style. The global model fitted on the pooled data without privacy is a reference that no private method can match in general. Results (Figure S9, left; 40 replicates per setting): at $\tau^2 = 0.05$, GHB 0.0166, Ditto-style 0.0198, pooled 0.0167, meta-analysis 0.0487, federated averaging 0.0617, local 0.0274. At $\tau^2 = 0.5$, GHB 0.0281, Ditto-style 0.0274, local 0.0273. One-shot federated averaging is weak here because it averages noisy releases without personalization; multi-round federated training with differentially private gradients would spend the budget over many rounds and is not compared.

Simulation 10: case-mix shift. As Simulation 8 at $\tau^2 = 0.05$, but each hospital draws its features from $\mathcal{N}(m_k, I)$ with $m_k \sim \mathcal{N}(0, s^2 I)$, while the de-shrinking still uses the public, unshifted feature distribution (Figure S9, right). At $s = 1$: GHB 0.0148, Ditto-style 0.0171, pooled 0.0198, local 0.0228.

Simulation 9: atypical hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$, with Gaussian site effects, Student-t_3 site effects of the same variance, or Gaussian effects plus one hospital at $u = 6\tau$, placed at the smallest or at the largest hospital. Three estimators: Gaussian GHB; robust GHB, which replaces the Gaussian prior $\mathcal{N}(\tilde{\mu}_k, \hat{\tau}^2 + M_k)$ by a Student-t_3 prior of the same variance (posterior mean by quadrature); and GHB with opt-out, where a hospital keeps its local estimate when $|\hat{\theta}_k - \tilde{\mu}_k|/\sqrt{v_k + \hat{\tau}^2 + M_k} > 3$. Both variants change only the hospital's own computation and cost no privacy. Mean relative risk: Gaussian effects 0.72 (robust 0.76), t_3 effects 0.69 (robust 0.65). The outlying hospital: 6.7 (robust 4.6, opt-out 4.4) when it is the smallest, 1.11 (robust 1.06) when it is the largest (Figure S10). With 16 patients, the standard error of the local estimate is as large as the deviation, so no test based on the hospital's own data can detect it reliably.

Simulation 11: a wrong anchor. The logistic world at $\tau^2 = 0.05$ and $\epsilon = 1$, with the anchor wrong by $\mathcal{N}(0, e^2 I)$ for e from 0 to 3. GHB uses the anchor with its uncertainty stated honestly (e), overconfidently (0.3 whatever e), overconfidently with a conflict check (the anchor is dropped when $|\mu_0 - \tilde{\mu}_k| > 3\sqrt{0.3^2 + M_k}$ in any coordinate), or not at all. The local model uses $\mathcal{N}(\mu_0, I)$ as its prior and degrades with the anchor (Figure S11). Honest uncertainty and the conflict check keep

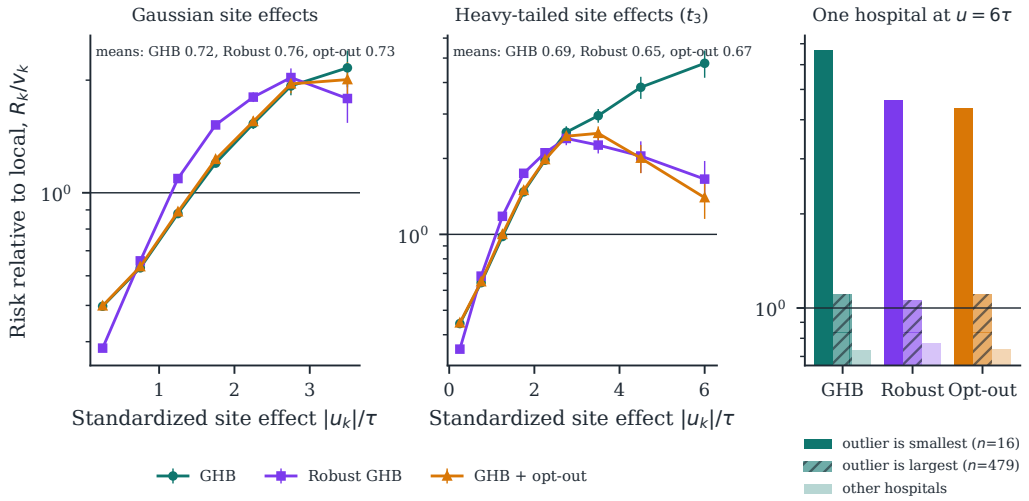


Fig. S10. Simulation 9. Left and centre: relative risk against the standardized site effect for Gaussian and heavy-tailed effects. Right: the outlying hospital and the others, when the outlier is the smallest or the largest hospital.

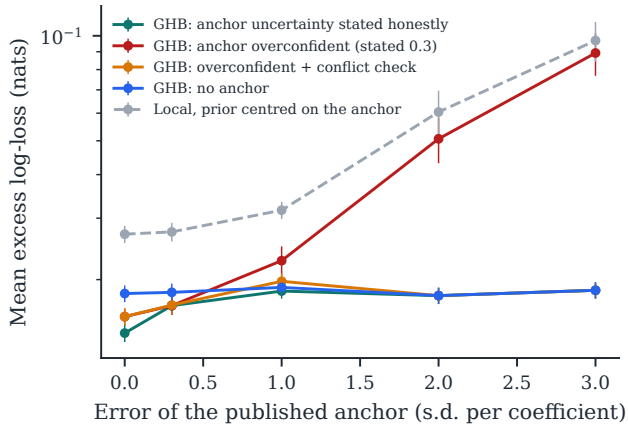


Fig. S11. Simulation 11. Mean excess log-loss against the error of the published anchor.

the excess log-loss at or below 0.0186 at every error, GHB without the anchor at or below 0.0190; an overconfident anchor two standard deviations off reaches 0.0506, and local estimation 0.0605.

Simulation 12: dishonest hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$. The one or three largest hospitals release $\mu + b\tau$ instead of their privatized estimate; sizes are public, so they cannot change their weights. Robust GHB obtains all releases by flooding, computes an inverse-variance weighted median and a MAD-based spread, drops releases more than three spreads away, and runs GHB on the rest; this is post-processing. Honest hospitals' mean relative risk (Figure S12): plain GHB never exceeds 1.00, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off; robust

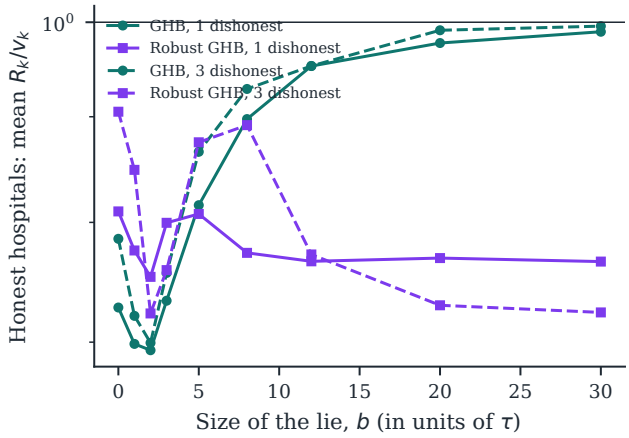


Fig. S12. Simulation 12. Honest hospitals' mean relative risk against the size of the lie, for one and three dishonest hospitals.

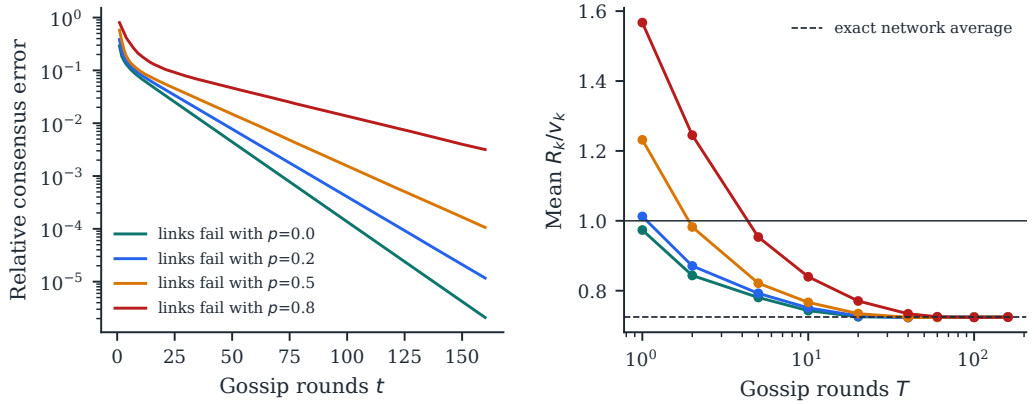


Fig. S13. Simulation 13. Left: consensus error against rounds. Right: mean relative risk of plug-in GHB against rounds, for four link-failure probabilities.

GHB keeps 0.72 against large lies, but costs 0.81 against 0.73 when nobody lies, because the screen sometimes drops honest outlying releases.

Simulation 13: link failures. The network of Simulation 2 with every link failing independently in each round with probability p; the Metropolis-Hastings weights are recomputed on the working links, so each round's matrix stays symmetric and doubly stochastic. With $p = 0.8$, the mean relative risk is 0.77 after 20 rounds and 0.73 after 60, against 0.72 with exact averaging (Figure S13).

Simulation 14: repeated releases. For a total budget of $\epsilon = 1$, $\delta = 10^{-5}$ spread over r refits on the same patients, the borrowed patients B after the last refit (closed form, network of Simulation 2) under basic composition ($(\epsilon/r, \delta/r)$ per release) and under zero-concentrated accounting (the Gaussian mechanism with noise s is ρ-zCDP with $\rho = \Delta^2/(2s^2)$; costs add; ρ-zCDP implies (ρ +

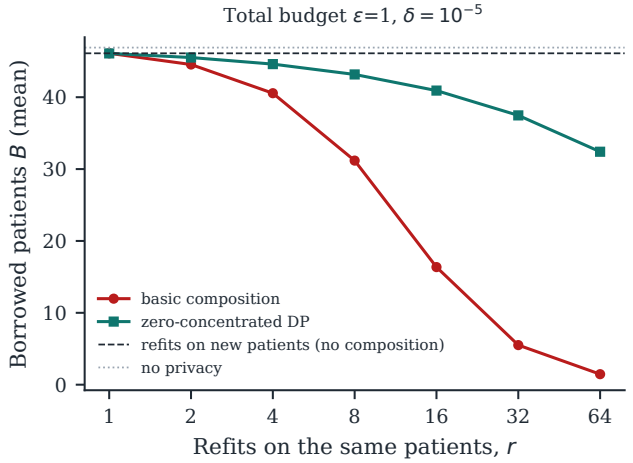


Fig. S14. Simulation 14. Borrowed patients after r refits on the same patients, for a fixed total budget.

$2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP). After 64 refits, $B = 1.5$ under basic composition and 32.4 under zCDP, against 46.1 for a single release (Figure S14).

S3 THE CODED CORPUS

Table S4 lists all 290 works of the corpus with their coding under the scheme of Table 3 of the paper. The same data are released as a CSV file, together with the scripts that compute every count in Sections 5 and 7 and draw Figures 7 and 8.

Table S4. The coded corpus (290 works), grouped by primary provenance channel and sorted by year. Sec.: secondary channel. Scarcity: classes of Figure 3 the work targets or documents. Type: Method, Theory, Study (single-dataset empirical study), Multi-site (training or validation across institutions), Resource (dataset, benchmark or infrastructure), Review, Evidence (documented failure or risk). General: method not specific to health data.

Work	Modality	Sec.	Scarcity	Type
P1 Pooling (102)				
DerSimonian and Laird [1986]	Tabular	P3	C1, T2	Method
Sweeney [2002]	Tabular		C2	Method
Xiao and Boyd [2004]	General		C1	Theory
Boyd et al. [2006]	General		C1	Theory
Dwork et al. [2006]	General		C2	Theory
Gentry [2009]	General		C2	Theory
Nedić and Ozdaglar [2009]	General		C1	Theory
Riley et al. [2010]	Tabular		C1, O2	Review
Overhage et al. [2012]	EHR		C2	Resource
Wu et al. [2012]	Tabular		C1	Method
Jiang et al. [2013]	Tabular		C1	Resource
Gaye et al. [2014]	Tabular		C1, C2	Resource
Hripcsak et al. [2015]	EHR		C1, C2	Resource
Lu et al. [2015]	Tabular		C1	Resource
Abadi et al. [2016]	General		C2	Method
Jochems et al. [2016]	Tabular		C1	Multi-site
Johnson et al. [2016]	EHR		C1, C2	Resource
Mandel et al. [2016]	EHR		C2	Resource
Blanchard et al. [2017]	General		C1	Method
Bonawitz et al. [2017]	General		C1, C2	Method
Jochems et al. [2017]	Tabular		C1	Multi-site
Kim et al. [2017]	EHR		C1	Method

Work	Modality	Sec.	Scarcity	Type
Lian et al. [2017]	General		C1	Theory
McMahan et al. [2017]	General		C1	Method
Mironov [2017]	General		C2	Theory
Papernot et al. [2017]	General	P4	C2	Method
Smith et al. [2017]	General	P3	C1, T2	Method
Vanhaesebrouck et al. [2017]	General	P3	C1, T2	Method
Bellet et al. [2018]	General	P3	C1, T2	Method
Brisimi et al. [2018]	EHR		C1	Method
Chang et al. [2018]	Imaging		C1	Multi-site
Cho et al. [2018]	Omics		C1, C2	Method
Gupta and Raskar [2018]	General		C1	Method
Lee et al. [2018]	EHR		C1, C2	Method
McMahan et al. [2018]	General		C1, C2	Method
Reps et al. [2018]	EHR		C2	Resource
Vepakomma et al. [2018]	Multimodal		C1, C2	Method
Zhao et al. [2018]	General		C1, T2	Method
Choudhury et al. [2019]	EHR		C1, C2	Method
Huang et al. [2019]	EHR		C1, T2	Multi-site
Jordan et al. [2019]	General		C1	Theory
Li et al. [2019]	Imaging		C1, C2	Method
Mohri et al. [2019]	General		C1, O3	Method
Poirot et al. [2019]	Imaging		C1, C2	Method
Raisaro et al. [2019]	Multimodal		C1, C2	Method
Roy et al. [2019]	Imaging		C1	Method
Sheller et al. [2019]	Imaging		C1	Method
Suchard et al. [2019]	EHR		C1, C2, O2	Multi-site
Yang et al. [2019]	General		C1	Review
Beyan et al. [2020]	Multimodal		C1, C2	Resource
Chen et al. [2020b]	Signals	P2	C1, T2	Method
Deist et al. [2020]	Tabular		C1	Multi-site
T. Dinh et al. [2020]	General	P3	C1, T2	Method
Duan et al. [2020b]	EHR		C1, O2	Method
Duan et al. [2020a]	EHR		C1, O2	Method
Fallah et al. [2020]	General	P2	C1, T2	Method
Ghosh et al. [2020]	General		C1, T2	Method
Hsieh et al. [2020]	General		C1, T2	Method
Kaissis et al. [2020]	Imaging		C1, C2	Review
Karimireddy et al. [2020]	General		C1, T2	Method
Koloskova et al. [2020]	General		C1	Theory
Li et al. [2020c]	General		C1, T2	Method
Li et al. [2020a]	Imaging	P2	C1, T2	Multi-site
Lin et al. [2020]	General	P4	C1, T2	Method
Rieke et al. [2020]	Multimodal		C1, C2	Review
Roth et al. [2020]	Imaging		C1, T2	Multi-site
Sheller et al. [2020]	Imaging		C1, O1	Multi-site
Wei et al. [2020]	General		C1, C2	Method
Bai et al. [2021]	Imaging		C1, T1	Multi-site
Collins et al. [2021]	General	P2	C1, T2	Method
Dayan et al. [2021]	Multimodal		C1, T1, O2	Multi-site
Dou et al. [2021]	Imaging		C1, T1	Multi-site
Froelicher et al. [2021]	Omics		C1, C2	Method
Guo et al. [2021]	Imaging		C1	Method
Huang et al. [2021a]	General		C1, T2	Method
Kairouz et al. [2021]	General		C1	Review
Kaissis et al. [2021]	Imaging		C1, C2	Method
Li et al. [2021a]	General	P3	C1, T2	Method
Li et al. [2021b]	General		C1, T2	Method
Liu et al. [2021]	Imaging	P6	C1, T2	Method
Marfoq et al. [2021]	General		C1, T2	Method
Sarma et al. [2021]	Imaging		C1	Multi-site
Tramèr and Boneh [2021]	General	P2	C2	Method
Vaid et al. [2021]	EHR		C1, T1	Multi-site
Warnat-Herresthal et al. [2021]	Multimodal		C1, O3	Multi-site
Xu et al. [2021]	Multimodal		C1	Review
Ziller et al. [2021]	Imaging		C2	Method
Adnan et al. [2022]	Imaging		C1, C2	Method
Cyffers and Bellet [2022]	General		C1, C2	Theory
Jiang et al. [2022]	Imaging		C1, T2	Method
Lu et al. [2022]	Imaging		C1, E1	Method
Nguyen et al. [2022]	Multimodal		C1	Review
Ogier du Terrail et al. [2022]	Multimodal		C1, T2	Resource
Pati et al. [2022]	Imaging		C1, O1	Multi-site
Saldanha et al. [2022]	Imaging		C1	Multi-site

Work	Modality	Sec.	Scarcity	Type
Tedeschini et al. [2022]	Imaging		C1	Method
Xu et al. [2022]	Imaging		C1, T2	Method
Kalra et al. [2023]	Imaging		C1, C2	Method
Lu et al. [2023]	General		C1	Method
Ogier du Terrail et al. [2023]	Imaging		C1, O1	Multi-site
Tan et al. [2023]	General		C1, T2	Review
Yan et al. [2023]	Imaging	P4	C1, E1	Method
P2 Proxy (60)				
Baxter [2000]	General		O1, E1	Theory
Ben-David et al. [2010]	General		T2	Theory
Pan and Yang [2010]	General		E1, T2	Review
Choi et al. [2016]	EHR		E1, T2	Method
Gulshan et al. [2016]	Imaging		E1	Study
Maurer et al. [2016]	General		O1	Theory
Shin et al. [2016]	Imaging		E1	Study
Tajbakhsh et al. [2016]	Imaging		E1	Study
Esteva et al. [2017]	Imaging		E1	Study
Finn et al. [2017]	General		O1, E1	Method
Kamnitsas et al. [2017]	Imaging	P4	T2	Method
Snell et al. [2017]	General		O1, E1	Method
Alsentzer et al. [2019]	Text		E1	Method
Chen et al. [2019]	Imaging		E1	Method
Gurovich et al. [2019]	Imaging		O1	Method
Prabhu et al. [2019]	Imaging		O1, O3	Method
Raghu et al. [2019]	Imaging		E1	Study
Zhang et al. [2019]	EHR		O1, E1	Method
Brown et al. [2020]	General		E1	Method
Lee et al. [2020]	Text		E1	Method
Li et al. [2020b]	EHR		E1, E3	Method
Mahajan et al. [2020]	Imaging		O1	Method
Roy et al. [2020]	Imaging		E1, O1	Method
Tripuraneni et al. [2020]	General		E1	Theory
Wang et al. [2020]	General		O1, E1	Review
Azizi et al. [2021]	Imaging	P4	E1, T2	Method
Du et al. [2021]	General		E1	Theory
Huang et al. [2021b]	Imaging	P4	E1	Method
Radford et al. [2021]	General		E1	Method
Rasmy et al. [2021]	EHR		E1, O2	Method
Sowrirajan et al. [2021]	Imaging	P4	E1	Method
Steinberg et al. [2021]	EHR		E1	Method
Zhou et al. [2021]	Imaging	P4	E1	Method
Agrawal et al. [2022]	Text		E1	Method
Guan and Liu [2022]	Imaging		T2	Review
Hospedales et al. [2022]	General		O1, E1	Review
Hsieh et al. [2022]	Imaging		O1	Method
Hu et al. [2022]	General		E1	Method
Mei et al. [2022]	Imaging		E1	Resource
Tiu et al. [2022]	Imaging	P4	E1	Method
Wang et al. [2022]	Imaging	P4	E1	Method
Zhang et al. [2022]	Imaging	P4	E1	Method
Banerjee et al. [2023]	Multimodal	P3	O1	Review
Hegselmann et al. [2023]	Tabular	P3	E1, O2	Method
Hollmann et al. [2023]	Tabular	P3	E3, O2	Method
Jiang et al. [2023]	Text		E1	Method
Moor et al. [2023]	Multimodal		E1, O1	Review
Nori et al. [2023]	Text		E1	Study
Singhal et al. [2023]	Text		E1	Method
Theodoris et al. [2023]	Omics		E1, E3, O1	Method
Wornow et al. [2023]	EHR		E1, T2	Review
Zhou et al. [2023]	Imaging	P4	E1, T2	Method
Chen et al. [2024]	Imaging		E1, O1	Method
Cui et al. [2024]	Omics		E1, E3	Method
Lu et al. [2024]	Imaging		E1, O1	Method
Ma et al. [2024]	Imaging		E1	Method
Pai et al. [2024]	Imaging		E1, O1	Method
Vorontsov et al. [2024]	Imaging		E1, O1	Method
Xu et al. [2024]	Imaging		E1	Method
Hollmann et al. [2025]	Tabular	P3	E3, O2	Method
P3 Prior (25)				
Stein [1956]	General	P1	O1, C1	Theory
Efron and Morris [1973]	General	P1	O1, C1	Theory
Tibshirani [1996]	General		E3	Method

Work	Modality	Sec.	Scarcity	Type
Steyerberg et al. [2000]	Tabular		O2, E3	Study
Lake et al. [2015]	General		O1	Method
Choi et al. [2017a]	EHR		E3, O1	Method
Rotmensch et al. [2017]	EHR	P4	E3	Method
Ma et al. [2018]	EHR		E3, O1	Method
Arjovsky et al. [2019]	General		T1, T2	Method
Raissi et al. [2019]	General		O1, E3	Method
Shang et al. [2019]	EHR		E3	Method
Subbaswamy et al. [2019]	EHR		T2	Method
Castro et al. [2020]	Imaging		T2, E2	Review
Kissas et al. [2020]	Imaging		E1	Method
Lagergren et al. [2020]	Tabular		E1, E3	Method
Sahli Costabal et al. [2020]	Signals		O1, E1	Method
Van Calster et al. [2020]	Tabular		O2	Study
Gulrajani and Lopez-Paz [2021]	General		T2	Study
Karniadakis et al. [2021]	General		O1, E3	Review
Köhler et al. [2021]	Text		O1	Resource
Lu et al. [2021a]	Tabular		O1	Method
Riley et al. [2021]	Tabular		O2	Study
Schölkopf et al. [2021]	General		T2	Review
Zhang et al. [2021]	Multimodal		T2	Study
von Rueden et al. [2023]	General		O1, E3	Review
P4 Unlabeled (23)				
Schölkopf et al. [2012]	General		E1	Theory
Agarwal et al. [2016]	EHR		E1, E2	Method
Halpern et al. [2016]	EHR	P3	E1, E2	Method
Bai et al. [2017]	Imaging		E1	Method
Ratner et al. [2017]	General	P3	E1, E2	Method
Wang et al. [2017]	Imaging		E1, E2	Resource
Ilse et al. [2018]	Imaging		E1	Method
Campanella et al. [2019]	Imaging		E1	Method
Cheplygina et al. [2019]	Imaging	P2	E1	Review
Fries et al. [2019]	Imaging		E1	Method
Irvin et al. [2019]	Imaging		E1, E2	Resource
Wang et al. [2019]	Text	P3	E1	Method
Yu et al. [2019]	Imaging		E1	Method
Chaitanya et al. [2020]	Imaging	P2	E1	Method
Chen et al. [2020a]	General	P2	E1	Method
Ouyang et al. [2020]	Imaging	P2	E1, O1	Method
Sohn et al. [2020]	General		E1	Method
Tajbakhsh et al. [2020]	Imaging	P6	E1, E2	Review
Taleb et al. [2020]	Imaging	P2	E1	Method
Li et al. [2021c]	Imaging		E1	Method
Lu et al. [2021b]	Imaging		E1	Method
He et al. [2022]	General	P2	E1	Method
Krishnan et al. [2022]	Multimodal	P2	E1	Review
P5 Procurement (15)				
Peduzzi et al. [1996]	Tabular		O2	Theory
Settles [2009]	General		E1	Review
Figueroa et al. [2012]	Text		E1	Method
Cho et al. [2015]	Imaging		E1	Study
Gal et al. [2017]	General		E1	Method
Yang et al. [2017]	Imaging		E1	Method
Smailagic et al. [2018]	Imaging		E1	Method
Balki et al. [2019]	Imaging		E1, O1	Review
Ghorbani and Zou [2019]	General		C1	Method
Riley et al. [2019b]	Tabular		O2, E3	Theory
Riley et al. [2019a]	Tabular		E3	Theory
van Smeden et al. [2019]	Tabular	P3	O2	Theory
Riley et al. [2020]	Tabular		O2, E3	Theory
Budd et al. [2021]	Imaging		E1	Review
Nath et al. [2021]	Imaging		E1	Method
P6 Re-expression (32)				
Chawla et al. [2002]	Tabular		O2	Method
Goodfellow et al. [2014]	General		E1	Method
Choi et al. [2017b]	EHR		C2	Method
Bowles et al. [2018]	Imaging		E1	Method
Fernández et al. [2018]	Tabular		O2	Review
Frid-Adar et al. [2018]	Imaging		E1, O2	Study
Shin et al. [2018]	Imaging		E1, C2	Method
Walonoski et al. [2018]	EHR	P3	C2	Resource

Work	Modality	Sec.	Scarcity	Type
Zhang et al. [2018]	General		E1	Method
Baowaly et al. [2019]	EHR		C2	Method
Beaulieu-Jones et al. [2019]	Tabular	P1	C2	Method
Jordon et al. [2019]	Tabular	P1	C2	Method
Shorten and Khoshgoftaar [2019]	General		E1	Review
Xu et al. [2019]	Tabular		C2	Method
Yi et al. [2019]	Imaging		E1	Review
Zhao et al. [2019]	Imaging		E1	Method
Goncalves et al. [2020]	EHR		C2	Study
Ho et al. [2020]	General		E1	Method
Yoon et al. [2020]	EHR		C2	Method
Zhang et al. [2020]	EHR		C2	Method
Chaitanya et al. [2021]	Imaging	P4	E1	Method
Chen et al. [2021a]	Multimodal		C2, O1	Review
Chlap et al. [2021]	Imaging		E1	Review
Jordon et al. [2022]	General		C2	Review
Stadler et al. [2022]	Tabular		C2	Study
van den Goorbergh et al. [2022]	Tabular		O2	Study
Borisov et al. [2023]	Tabular	P2	C2	Method
Kazerouni et al. [2023]	Imaging		E1	Review
Khader et al. [2023]	Imaging		E1, C2	Method
Kotelnikov et al. [2023]	Tabular		C2	Method
Ktena et al. [2024]	Imaging		O3, T2	Method
Shumailov et al. [2024]	General		O3	Evidence
No remedy: documented scarcity, failure or risk (33)				
El Emam et al. [2011]	Tabular		C2	Review
Button et al. [2013]	General		O2	Evidence
Shokri et al. [2017]	General		C2	Evidence
Varoquaux [2018]	Imaging		O1, O2	Evidence
Zech et al. [2018]	Imaging		T2	Evidence
Bagdasaryan et al. [2019]	General		O3, C2	Evidence
Brasil et al. [2019]	Multimodal		O1	Review
Christodoulou et al. [2019]	Tabular		O2	Review
Jayaraman and Evans [2019]	General		C2	Evidence
Kim et al. [2019]	Imaging		T2	Review
Liu et al. [2019]	Imaging		T2	Review
Obermeyer et al. [2019]	EHR		O3	Evidence
Rocher et al. [2019]	Tabular		C2	Evidence
Schwarz et al. [2019]	Imaging		C2	Evidence
Zhu et al. [2019]	General		C1, C2	Evidence
Geiping et al. [2020]	General		C1, C2	Evidence
Nagendran et al. [2020]	Imaging		T2	Review
Nguengang Wakap et al. [2020]	Tabular		O1	Evidence
Oakden-Rayner et al. [2020]	Imaging		O3	Evidence
Schaefer et al. [2020]	Multimodal		O1	Review
Willemink et al. [2020]	Imaging		E1, E2, C2	Review
Wynants et al. [2020]	Multimodal		O2, T1	Evidence
Chen et al. [2021b]	Multimodal		O3	Review
Finlayson et al. [2021]	Multimodal		T1, T2	Evidence
Koh et al. [2021]	Multimodal		T2	Resource
Roberts et al. [2021]	Imaging		O2, T1	Evidence
Wong et al. [2021]	EHR		T2	Evidence
Carlini et al. [2022]	General		C2	Evidence
Gichoya et al. [2022]	Imaging		O3, T2	Evidence
Packhäuser et al. [2022]	Imaging		C2	Evidence
Varoquaux and Cheplygina [2022]	Imaging		O1, T2	Evidence
Yu et al. [2022]	Imaging		T2	Review
Hatamizadeh et al. [2023]	Imaging		C1, C2	Evidence

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